

Quiz 2

Question

In each case below provide an example of a sequence of rational numbers satisfying the required property.

- A sequence that converges in $|\cdot|_2$ and in $|\cdot|_5$ but not in $|\cdot|_\infty$.
- A sequence that converges in $|\cdot|_2$ and in $|\cdot|_\infty$ but not in $|\cdot|_5$.
- A sequence that converges in $|\cdot|_5$ but not in $|\cdot|_2$.
- A sequence that converges in *all* the absolute values on $\mathbb{Q}$.

Solution.

- The sequence $(10^n)_{n=0}^\infty$ converges in $|\cdot|_2$ and $|\cdot|_5$ to 0. However, it does not converge in $|\cdot|_\infty$. In fact, it diverges to infinity.
- The sequence $((2/5)^n)_{n=0}^\infty$ converges in $|\cdot|_\infty$ and $|\cdot|_2$ to 0. However, it does not converge in $|\cdot|_5$ as it is unbounded.
- The sequence $((5/2)^n)_{n=0}^\infty$ converges in $|\cdot|_5$ to 0. However it is unbounded in $|\cdot|_2$.
- Any constant sequence converges in *all* absolute values on $\mathbb{Q}$.

Non-questions

- Let $f : \mathbb{N} \rightarrow \mathbb{Q}$ be surjection (such as an enumeration of the countable set of the rational numbers) and define $x_n = f(n)$.

Show that this sequence contains the following sub-sequences:

- An increasing sequence of positive integers.
- For each prime p , a sub-sequence of the form p^{-n_k} where $(n_k)_k$ is an increasing sequence of integers.

Conclude that x_n does not converge in *any* absolute value on the rational numbers.

- Let $(p_n)_n$ be the sequence of primes, define $N_k = \prod_{i=1}^k p_i^k$ and $x_k = N_k/(1 + N_k)$.

Show that x_k converges in all absolute values on $\mathbb{Q}$.